

Spectral Energy Distribution Analyzer (SEDA) for Forward Modeling and Empirical Analysis of Ultracool Objects

Genaro Suárez

American Museum of Natural History



Collaborators:

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Sherelyn Alejandro Merchan (CUNY)

Austin Rothermich (CUNY)

Melanie Rowland (AMNH)

Niall Whiteford (AMNH)

Jolie L'Heureux (CUNY)

et al.



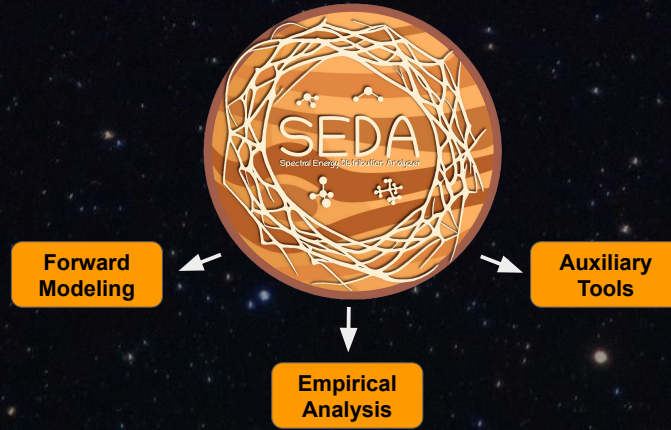
III New York Area Exoplanets Meeting

AMNH, New York. May 15, 2026

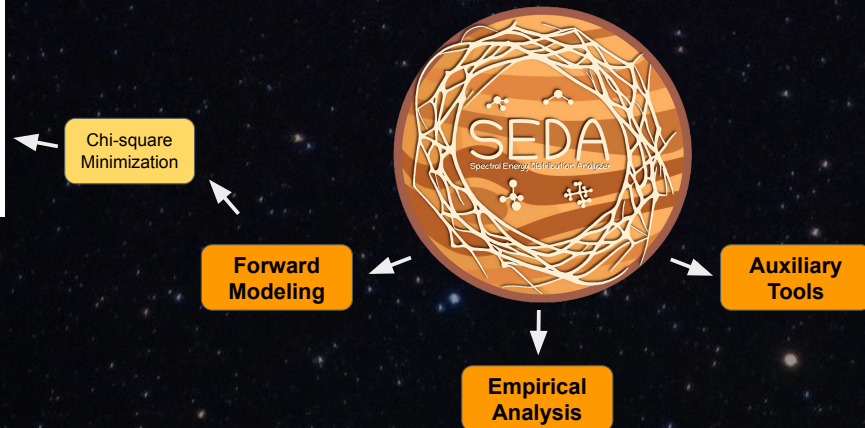
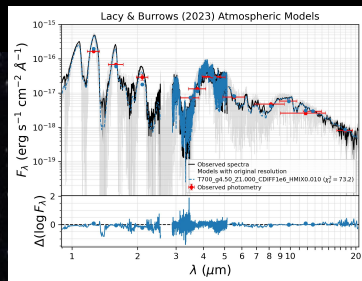
Overview



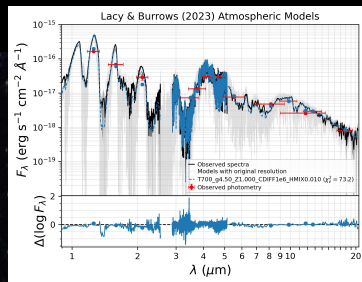
Overview



Overview



Overview



Chi-square
Minimization

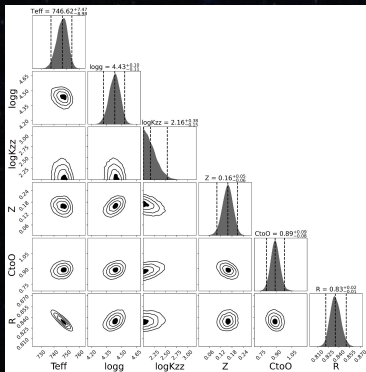
Forward
Modeling

Nested
Sampling

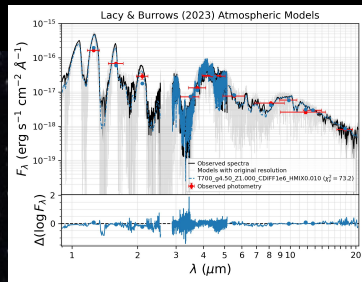


Auxiliary
Tools

Empirical
Analysis



Overview



Chi-square
Minimization

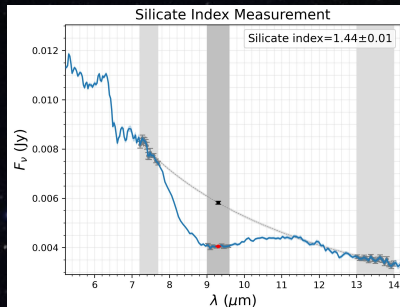
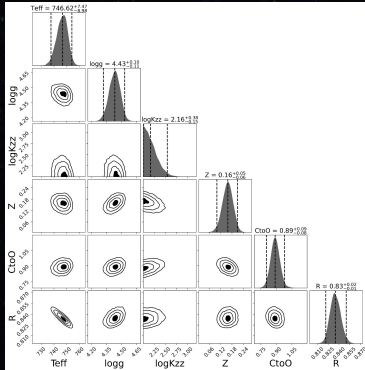
Forward
Modeling

Auxiliary
Tools

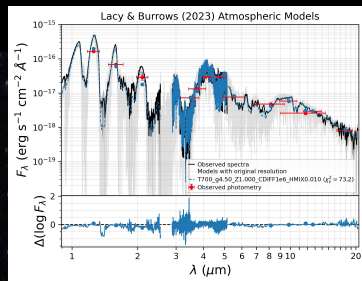
Empirical
Analysis

Nested
Sampling

Spectral
Indices



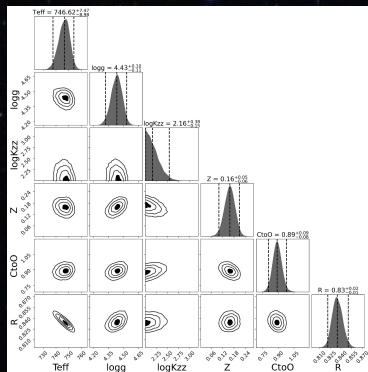
Overview



Chi-square
Minimization

Forward
Modeling

Nested
Sampling

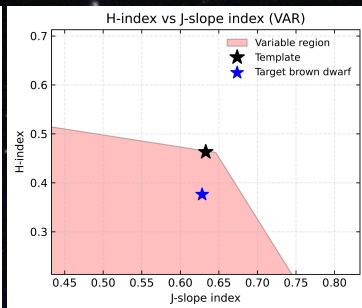
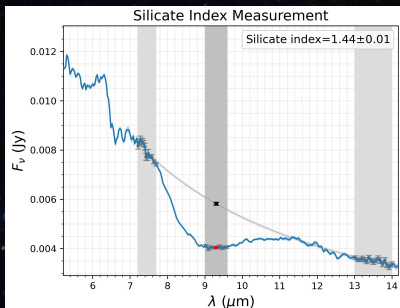


Auxiliary
Tools

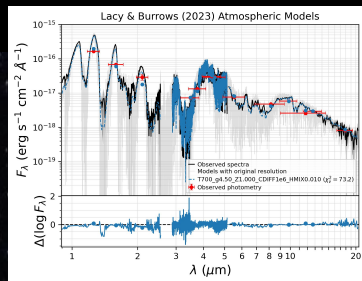
Empirical
Analysis

Spectral
Indices

Variability



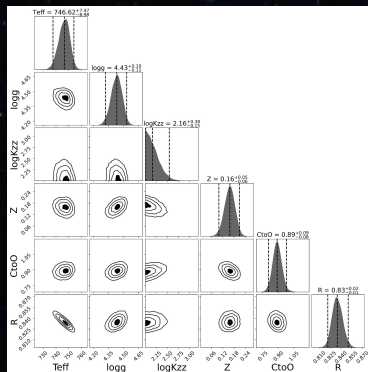
Overview



Chi-square
Minimization

Forward
Modeling

Nested
Sampling



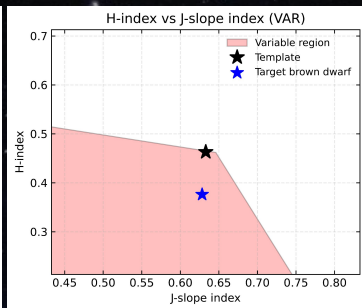
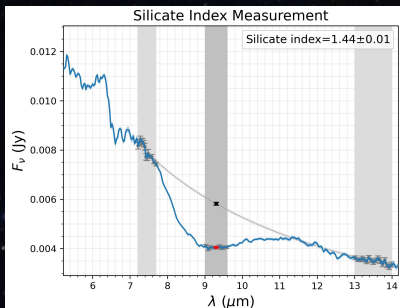
Auxiliary
Tools

Empirical
Analysis

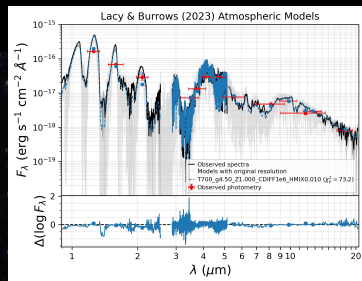
Spectral
Indices

Variability

(Semi)
empirical
Parameters



Overview



Chi-square Minimization

Forward Modeling



Auxiliary Tools

Empirical Analysis

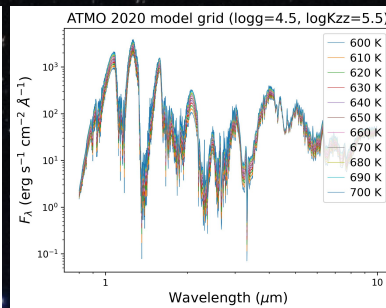
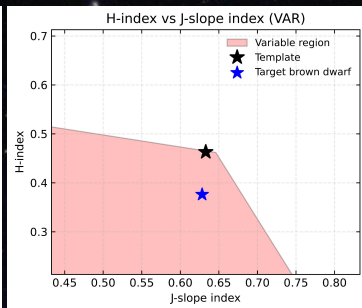
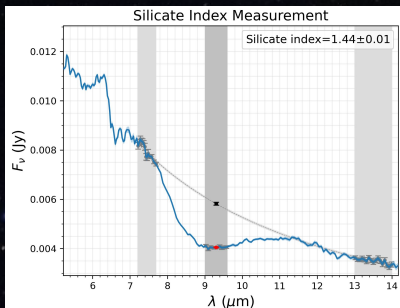
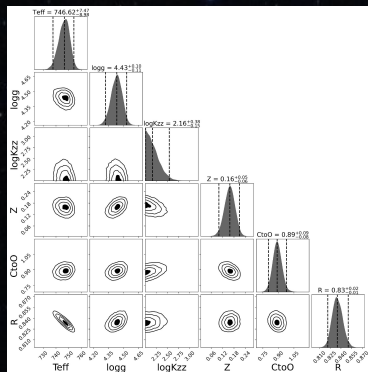
Nested Sampling

Spectral Indices

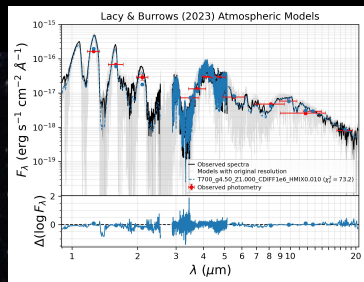
Variability

(Semi) empirical Parameters

Model Interpolator



Overview



Chi-square
Minimization

Forward
Modeling



Auxiliary
Tools

Empirical
Analysis

Synthetic
Photometry

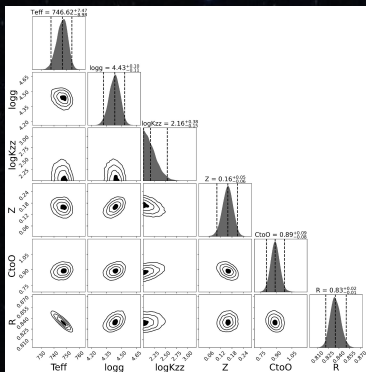
Nested
Sampling

Model
Interpolator

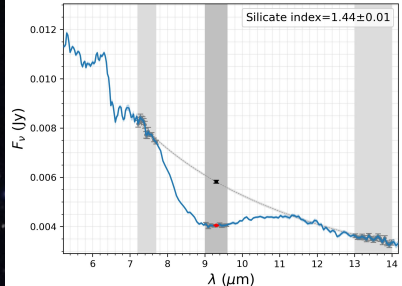
Spectral
Indices

Variability

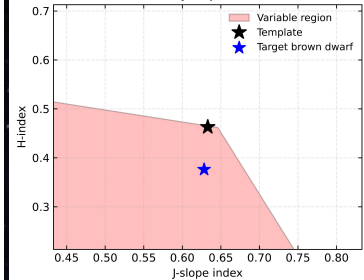
(Semi)
empirical
Parameters



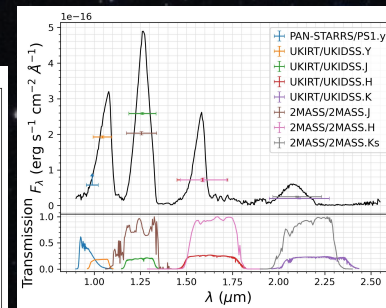
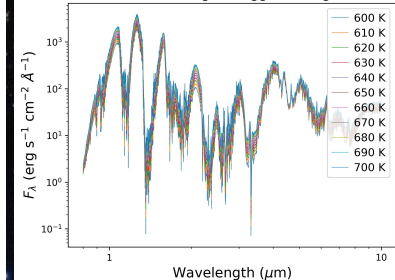
Silicate Index Measurement



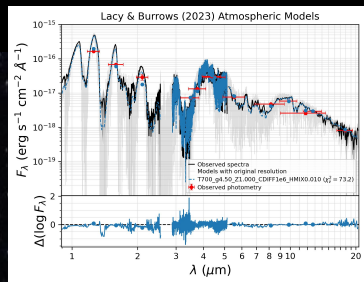
H-index vs J-slope index (VAR)



ATMO 2020 model grid (logg=4.5, logKzz=5.5)



Overview



Chi-square
Minimization

Forward
Modeling



Auxiliary
Tools

Model
Examiner

Nested
Sampling

Empirical
Analysis

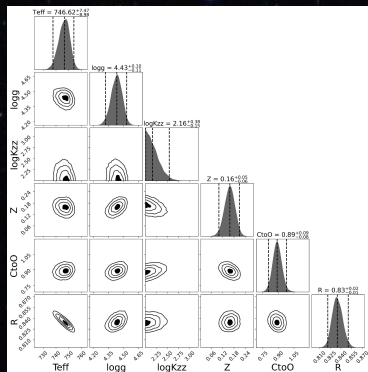
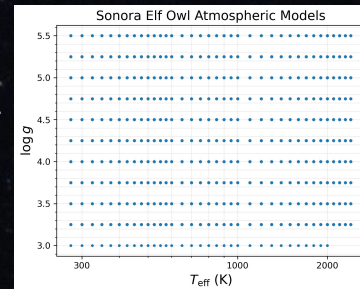
Synthetic
Photometry

Spectral
Indices

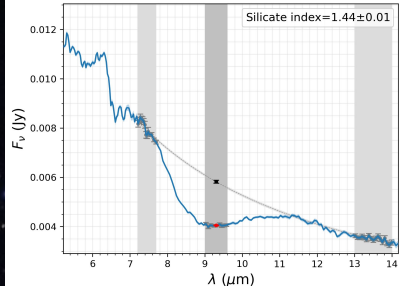
Variability

(Semi)
empirical
Parameters

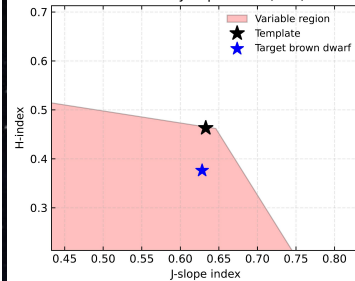
Model
Interpolator



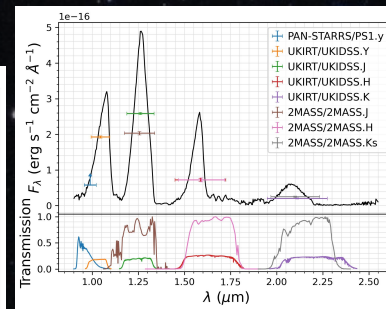
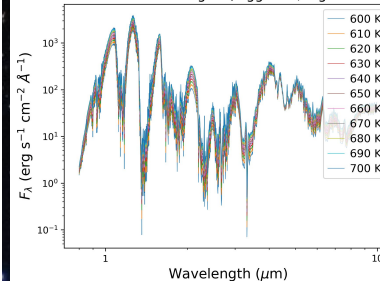
Silicate Index Measurement



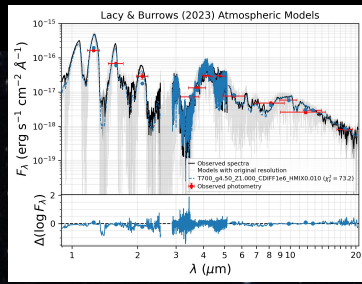
H-index vs J-slope index (VAR)



ATMO 2020 model grid (logg=4.5, logKzz=5.5)



Overview



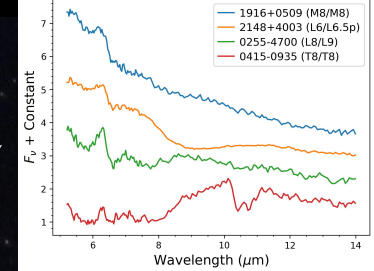
Chi-square
Minimization

Forward
Modeling

Auxiliary
Tools

Archival
Spectra

Spitzer IRS Library

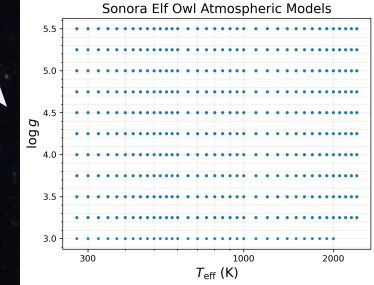


Nested
Sampling

Empirical
Analysis

Model
Examiner

Synthetic
Photometry

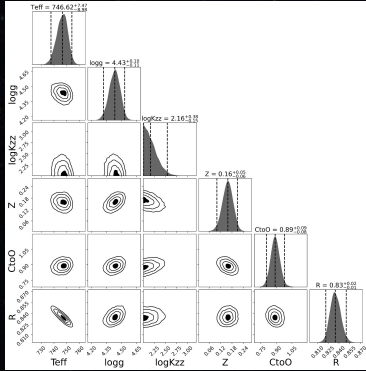


Spectral
Indices

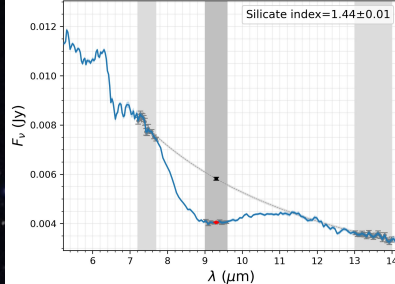
Variability

(Semi)
empirical
Parameters

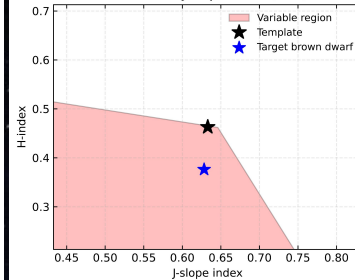
Model
Interpolator



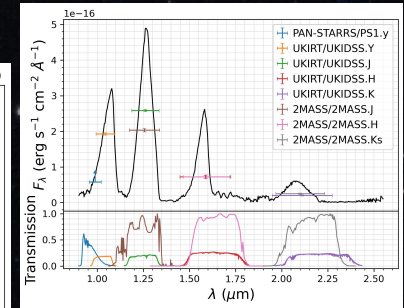
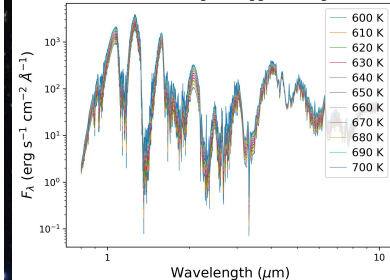
Silicate Index Measurement



H-index vs J-slope index (VAR)



ATMO 2020 model grid ($\log g=4.5$, $\log K_{\text{zz}}=5.5$)



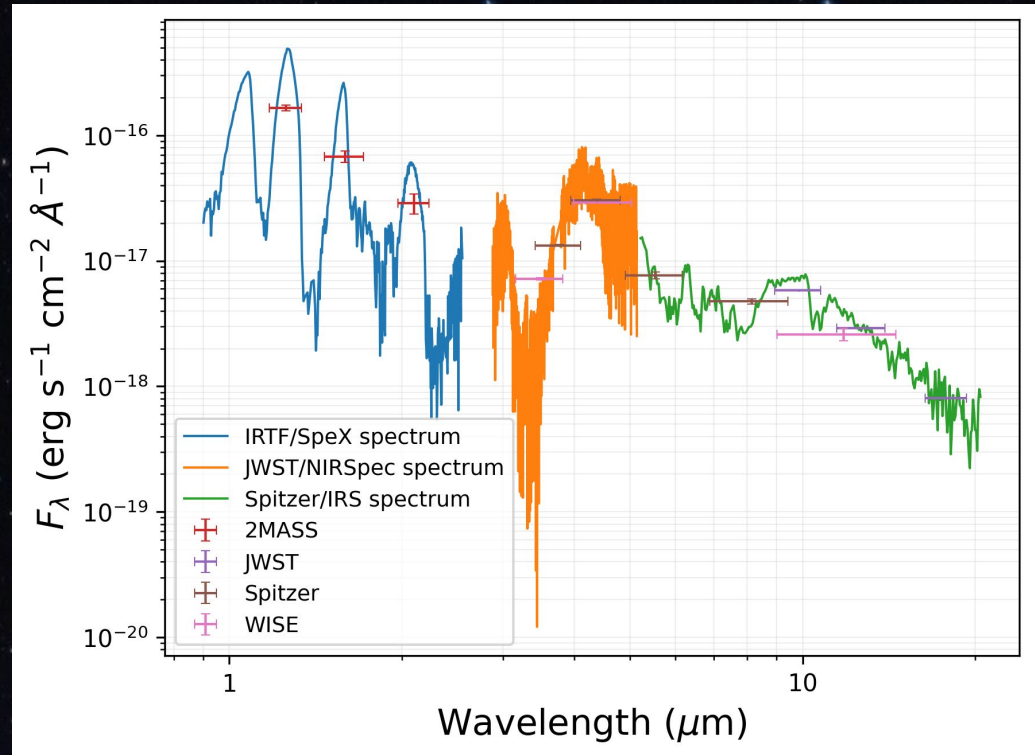
Forward Modeling Workflow



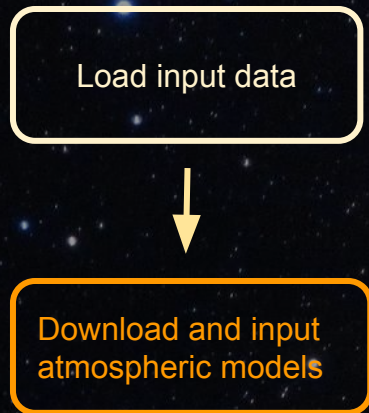
Forward Modeling Workflow

Load input data

Observed SED for the T8 (~750 K)
brown dwarf J0415 (Alejandro Merchan+2025)



Forward Modeling Workflow



Available atmospheric models:

- Lacy et al. (2026, in prep.)
- Sonora Elf Owl models (Mukherjee et al. 2024)
- Sonora Diamondback (Morley et al. 2024)
- SAND (Alvarado et al. 2024)
- Lacy & Burrows (2023)
- Sonora Cholla (Karalidi et al. 2021)
- Sonora Bobcat (Marley et al. 2021)
- ATMO 2020 (Phillips et al. 2020)
- BT-Settl (Allard et al. 2012)
- Saumon & Marley (2008)

Tutorial available for ingesting new model grids.

Forward Modeling Workflow

Load input data

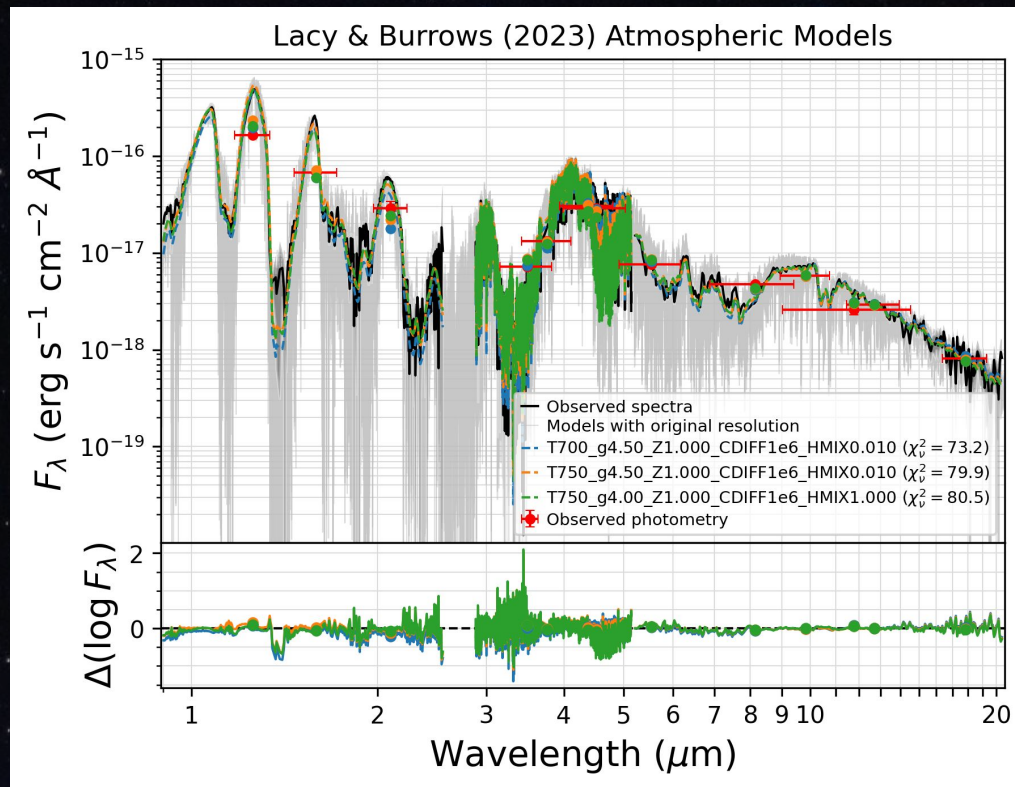


Download and input
atmospheric models

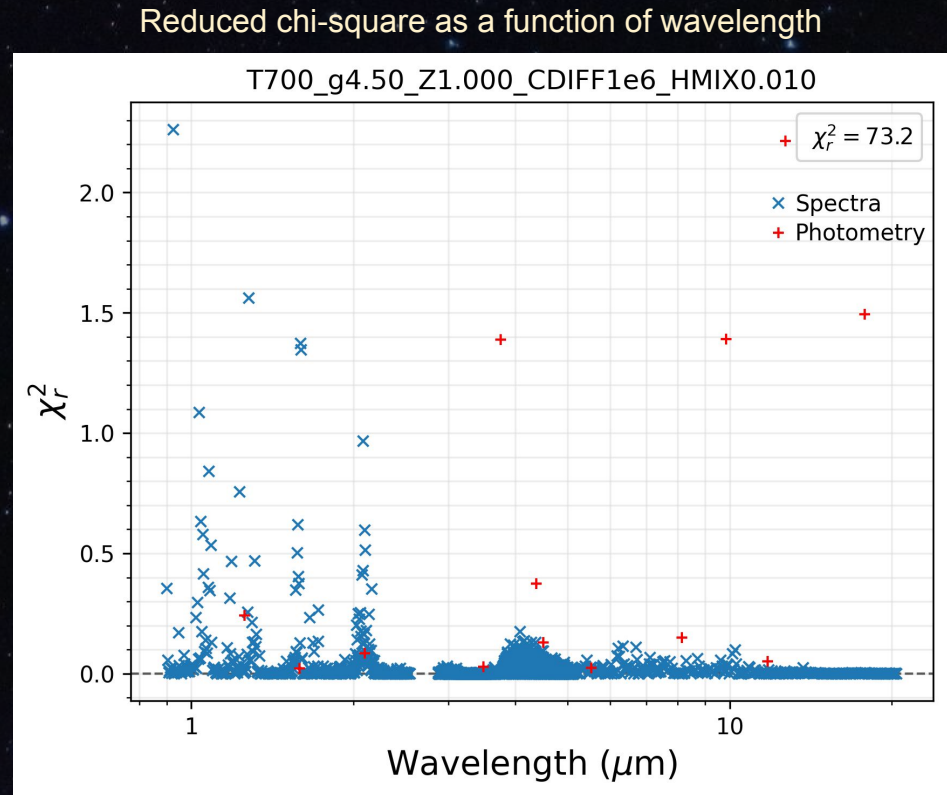
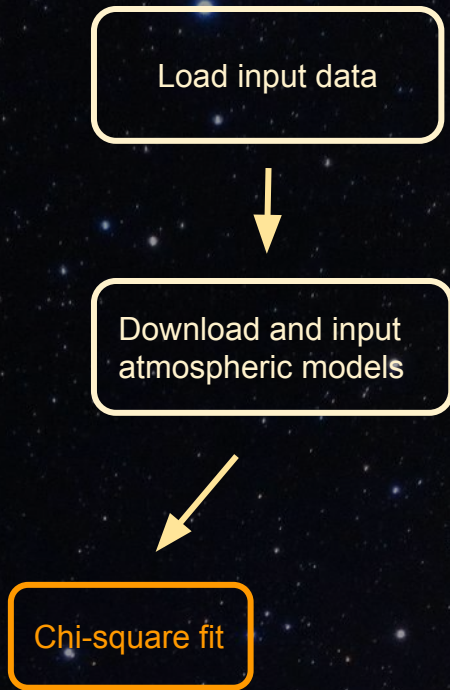


Chi-square fit

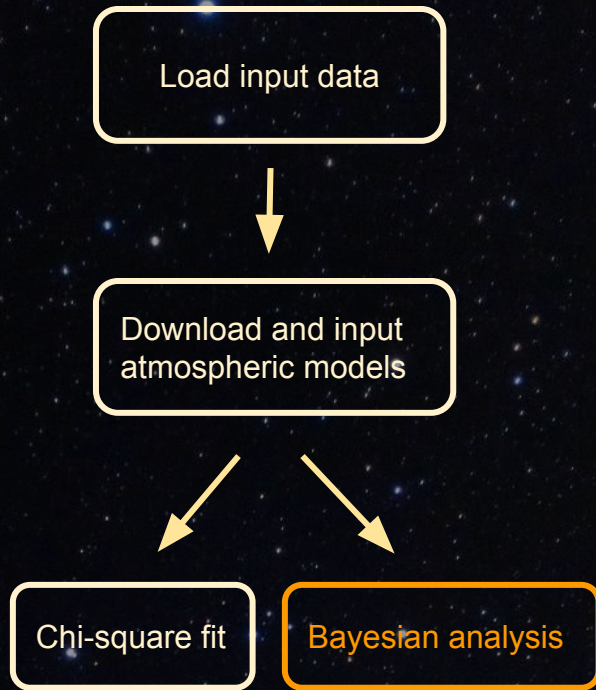
Best three model fits



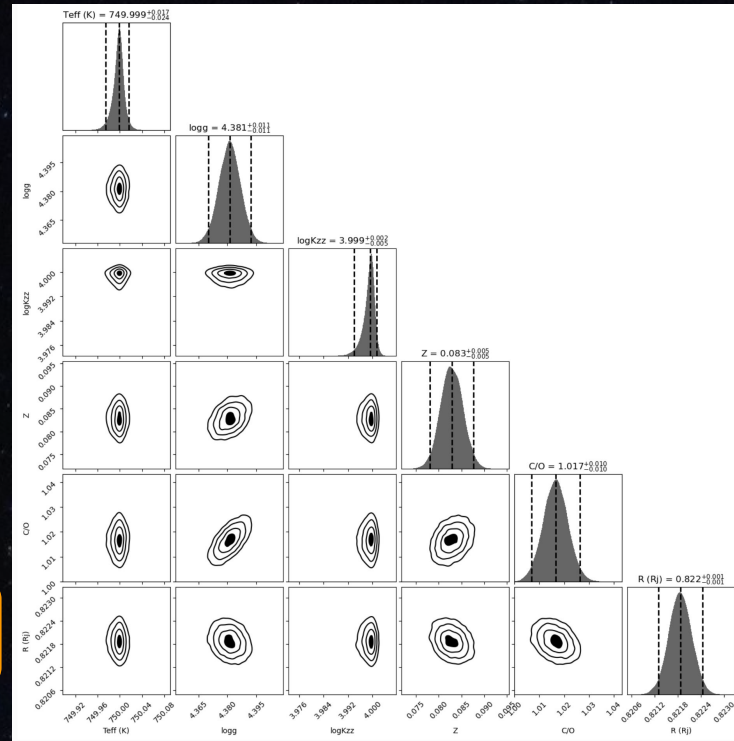
Forward Modeling Workflow



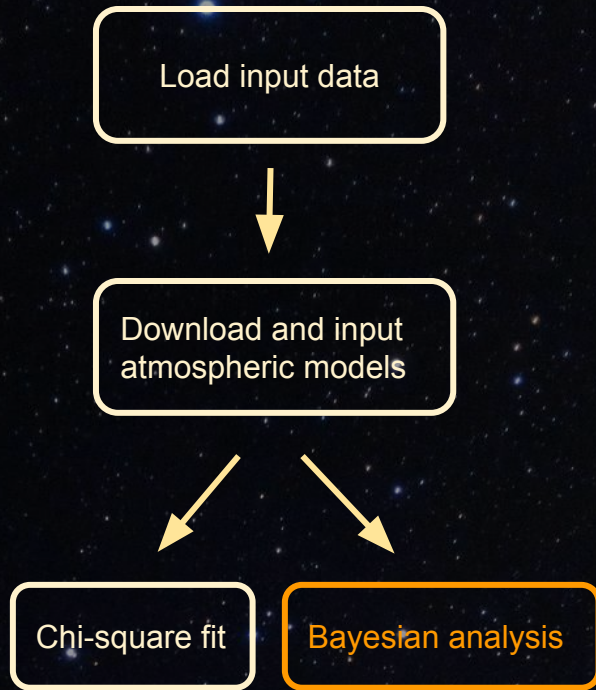
Forward Modeling Workflow



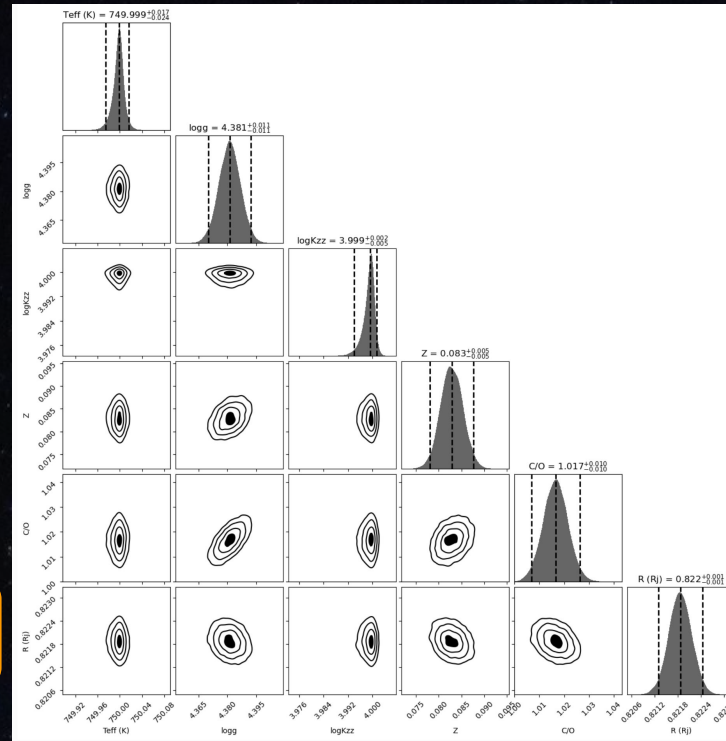
Corner plot



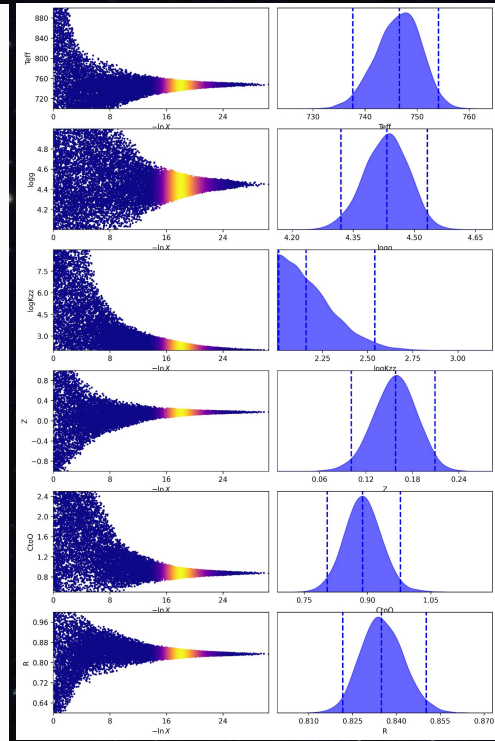
Forward Modeling Workflow



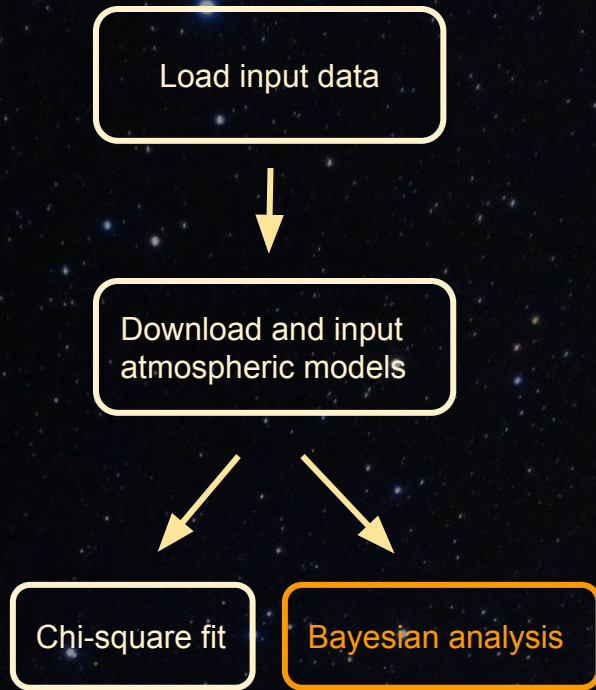
Corner plot



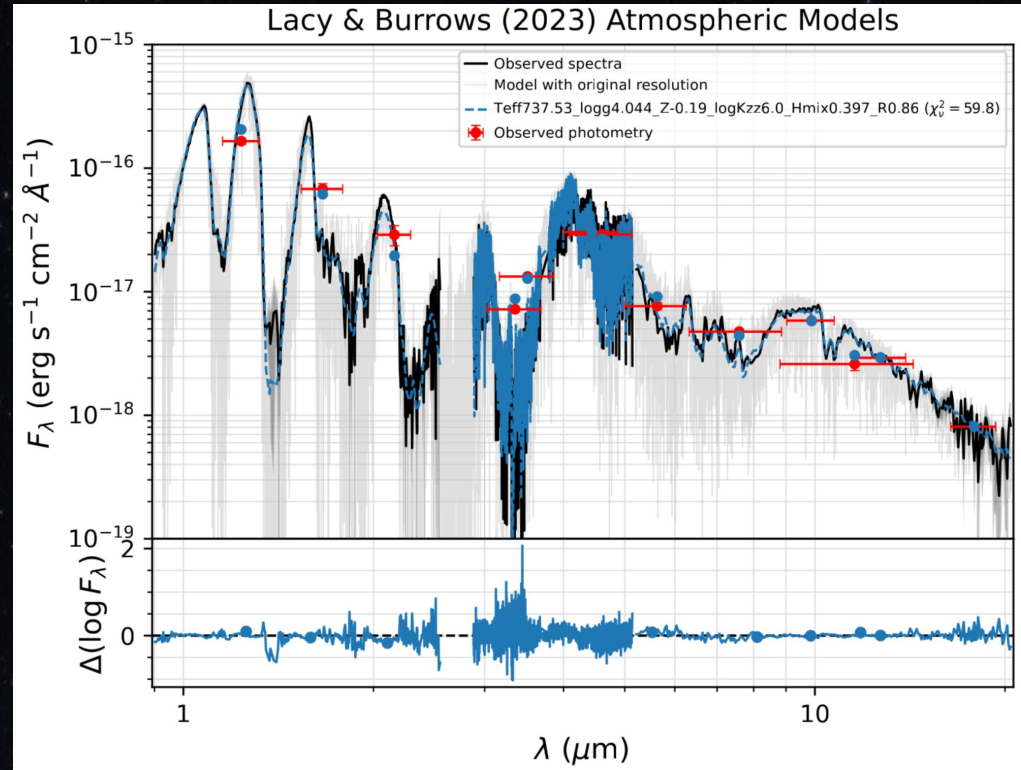
Trace plot



Forward Modeling Workflow



Model spectrum with the median posteriors



Suárez et al. (2026, in prep.)

Principal Tools



Principal Tools

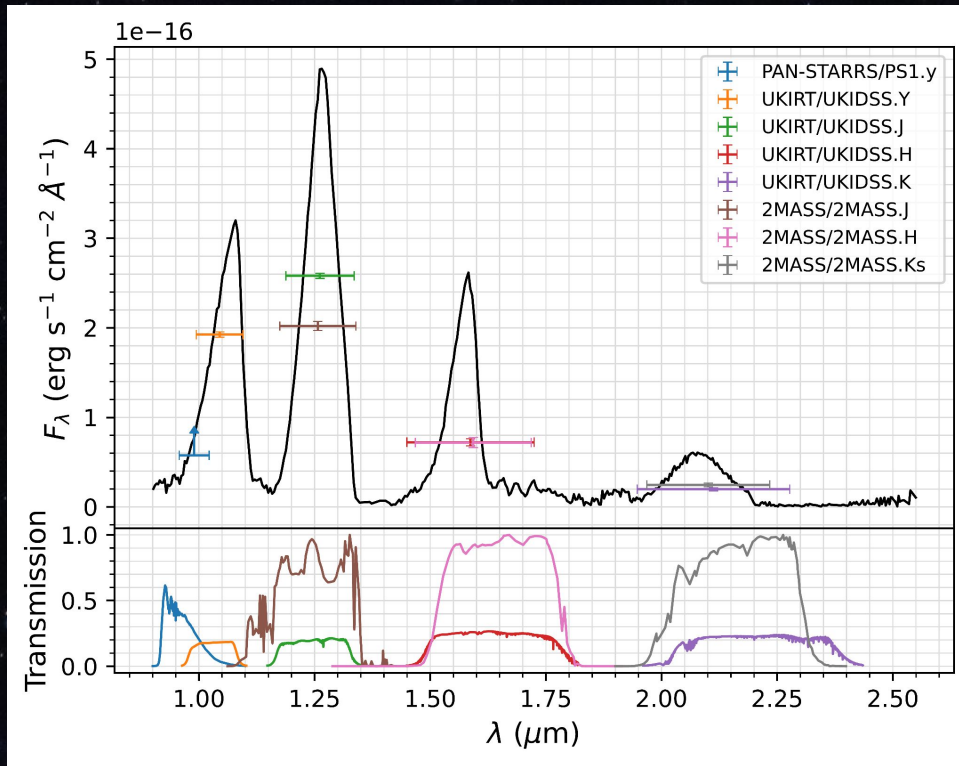
- Synthetic photometry for any SVO filters

Filter Profile Service
Carlos Rodrigo
A repository of filter information for the VO.

VO Service Browse Search News Help-Desk Search text ~ Search (?)

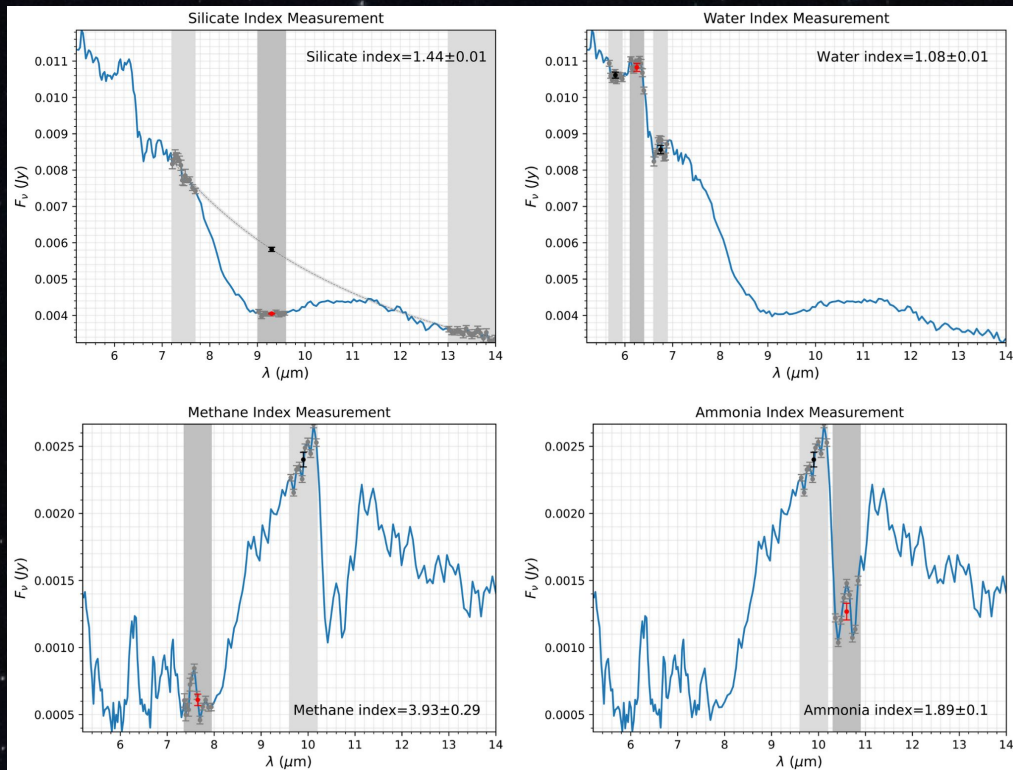
Astronomy (7905) Planetary science (616) Earth Obs. (2452)

2MASS	AAG	ADIOS	AKARI	Alakatsi	Alsat1	AFEX	ARO	Aqua	ARCHERPS	ARGO	AstroSat	Beijing1	Bepi-Colombo	BICEP
BLAST	BOK	BOOMERANG	CAHA	Cameras	Cassini	CASTOR	CFHT	CHEOPS	Clementine	CNO-SAT	COBE	COMS	Contour	Corot
COSMOSOMAS	CSST	CSTAR	CTIO	Dawn	DeepImpact	DENIS	DOT	DISCOVER	Enviat	ERBS	EROS	ERS	ESO	Euclid
ExoMars	FengYun	Flock	FLWO	GAIA	GALEX	Galileo	GCOM-C	GCPD	Gemini	Generic	Geneva	GeoEye	Giotto	GOES
GOTO	GTC	Hayabusa2	HCT	Herschel	Hinawari	Hipparcos	HST	IACOB	IKONOS	ING	INSAT	Indisight	INT	Integral
IRAM	IRAS	IRS	IRSP	IRTF	ISO	IUE	JCHT	JPSS	JWST	Keck	Kepler	KOMPSTAT	KPN0	Landiat
LacCumbres	LaSilla	LBT	LCO	LICK	Liverpool	LMT	LRO	LSST	LYRA	Mariner10	Mars2020	MAXIMA	McD	MER
Mercator	Messenger	Meteorat	METOP	MEX	Misc	MKG	MMT	MOA	MOH	MOIST	MRO	MSX	MT	NAOC
NEAR	NewHorizons	NigerSat1	NIMBUS	NIRT	NOAA	NOAO	NOT	OAF	OAJ	DAN-SPM	OAO	Odyssey	OHF	OLIMPO
OSIRIS-REX	OSN	OVID	P200	Pallat	PAN-STARRS	Paranal	Parasol	Patfinder	PLANCK	PRIMO	PRIMO	Quickline	QUIET	QUOTE
RapidEye	Roman	Rosetta	SALT	Sao	Scorpio	Seafar	Selene	Sentinel	SEOSAT	SlyMapper	SlySat	SLOAN	SMART1	SOPRA
SOHO	SolarOrbiter	Speical	SPECULOS	SPIDER	Spotter	SPOT	SPT	SSOT	Stardust	STELLA	Subaru	Swift	TAUVEX	TCS
TD1	Terra	TESS	TIROS-N	TJO	TNG	TNO	TNT	Tophat	TRMM	TYCHO	UK-DMC	UKIRT	VATT	VenusExpress
Viking	Voyager	WASP	WFIRST	WHT	WISE	WYRN	WMAP	Worldview	XMM	ZYuan				



Principal Tools

- Synthetic photometry for any SVO filters
- Measure the depth of spectral features

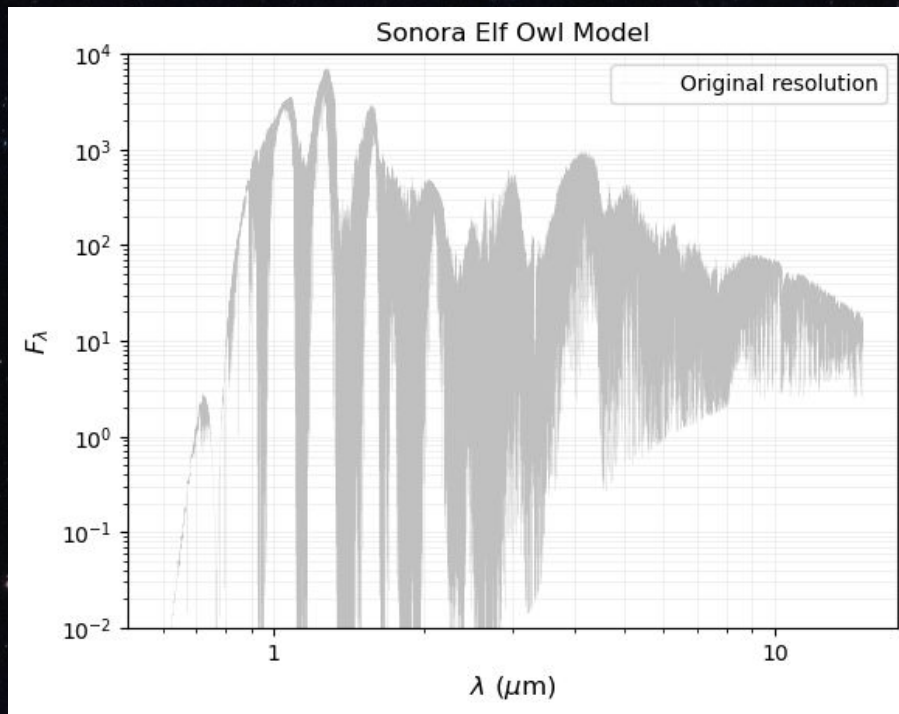


Principal Tools

- Synthetic photometry for any SVO filters
- Measure the depth of spectral features
- Generate model spectra with parameters with the grid coverage

Sonora Elf Owl model:

$T_{\text{eff}}=789$ K, $\log g=4.38$, $\log K_{\text{zz}}=5.21$, $Z=-0.28$, $\text{C/O}=1.13$

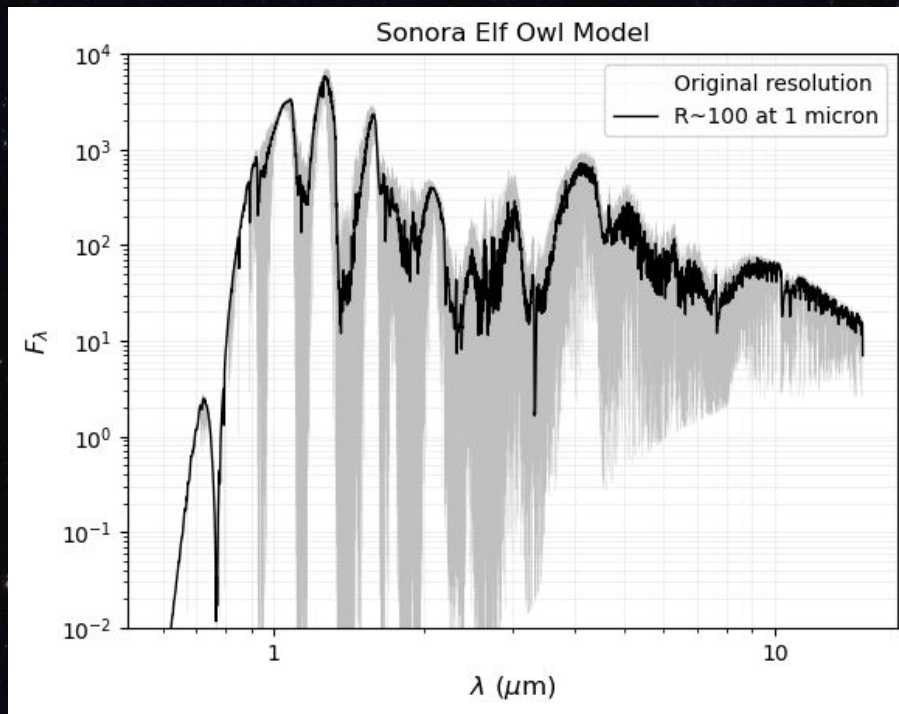


Principal Tools

- Synthetic photometry for any SVO filters
- Measure the depth of spectral features
- Generate model spectra with parameters with the grid coverage
- Convolve spectra to a desired resolution

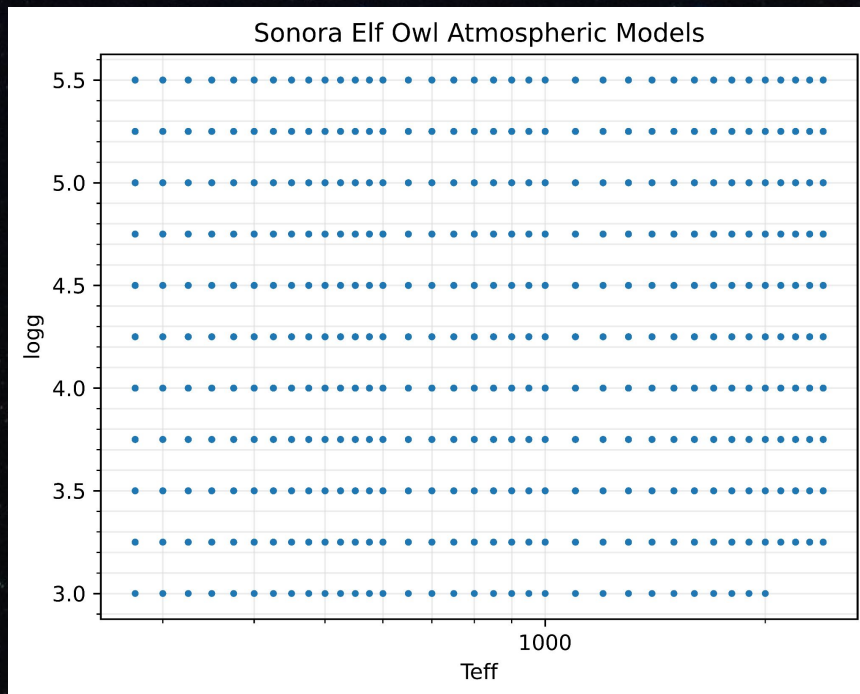
Sonora Elf Owl model:

$T_{\text{eff}}=789$ K, $\log g=4.38$, $\log K_{\text{zz}}=5.21$, $Z=-0.28$, $\text{C/O}=1.13$



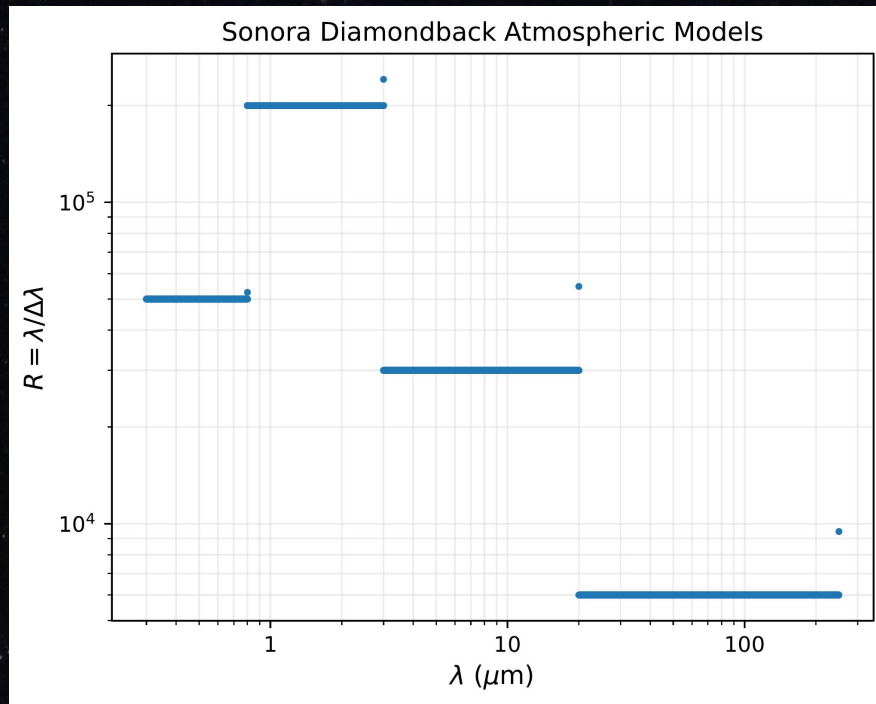
Principal Tools

- Synthetic photometry for any SVO filters
- Measure the depth of spectral features
- Generate model spectra with parameters with the grid coverage
- Convolve spectra to a desired resolution
- Inspect model parameters coverage and resolution



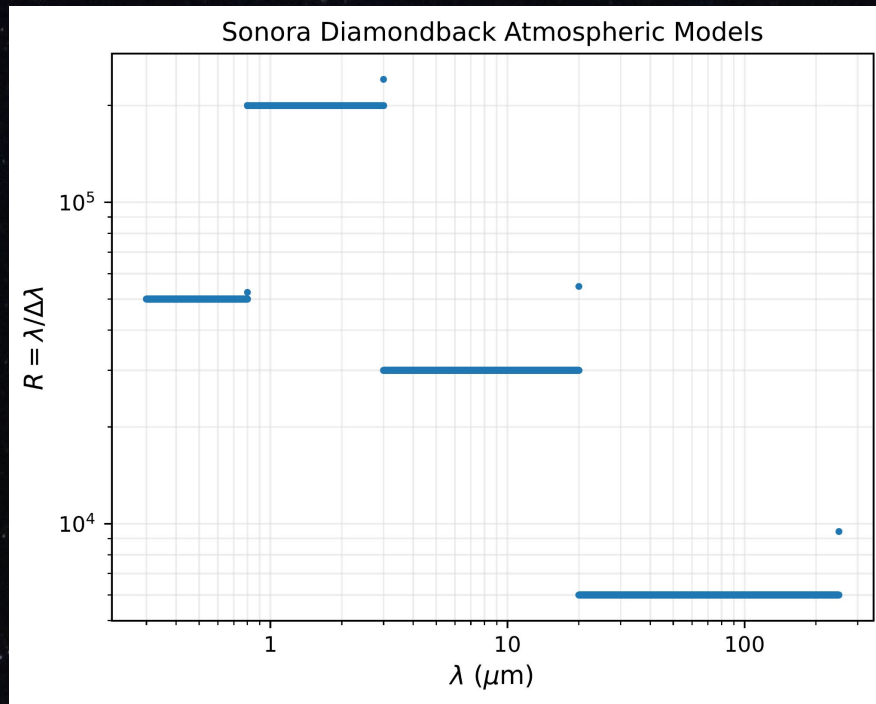
Principal Tools

- Synthetic photometry for any SVO filters
- Measure the depth of spectral features
- Generate model spectra with parameters with the grid coverage
- Convolve spectra to a desired resolution
- Inspect model parameters coverage and resolution



Principal Tools

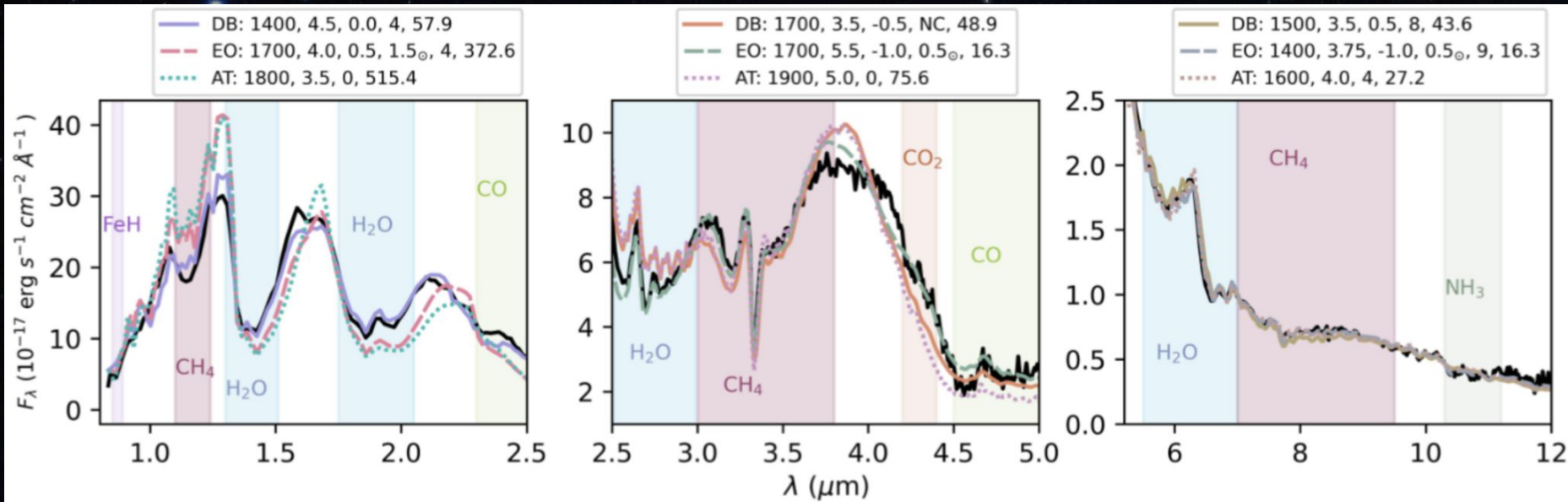
- Synthetic photometry for any SVO filters
- Measure the depth of spectral features
- Generate model spectra with parameters with the grid coverage
- Convolve spectra to a desired resolution
- Inspect model parameters coverage and resolution
- Convert fluxes into magnitudes and vice versa



Application 1: Modeling Multiple Wavelength Regions



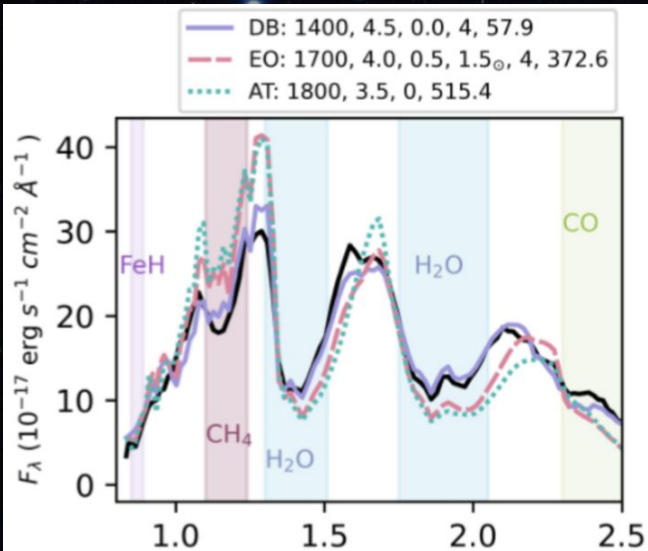
Application 1: Modeling Multiple Wavelength Regions



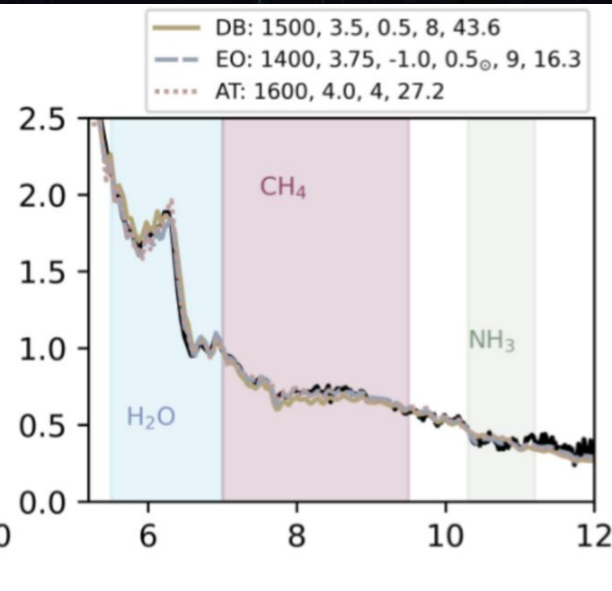
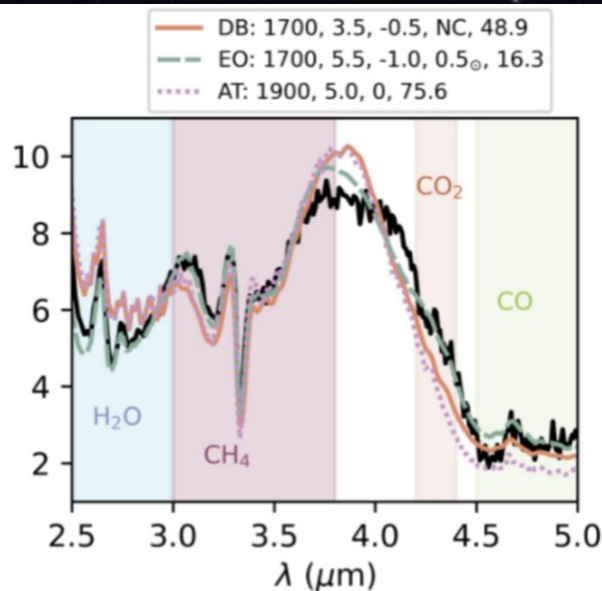
Rothermich et al. (2026, submitted)

Application 1: Modeling Multiple Wavelength Regions

Best fit: Cloudy models



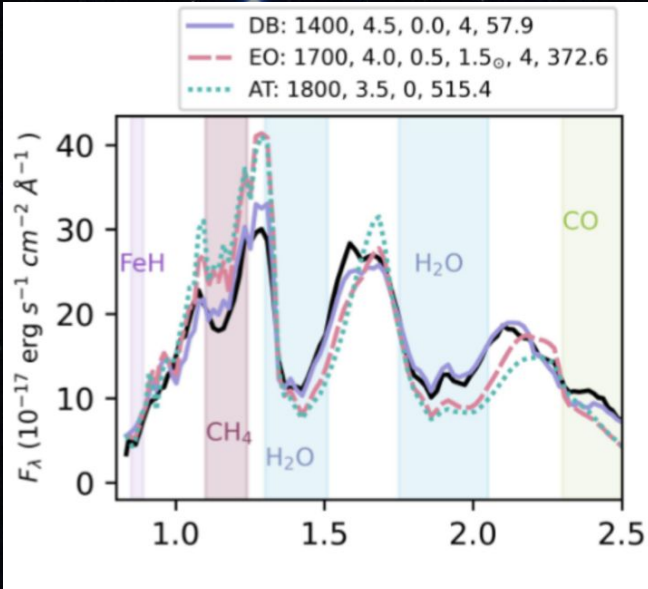
Best fit: Cloud-free models



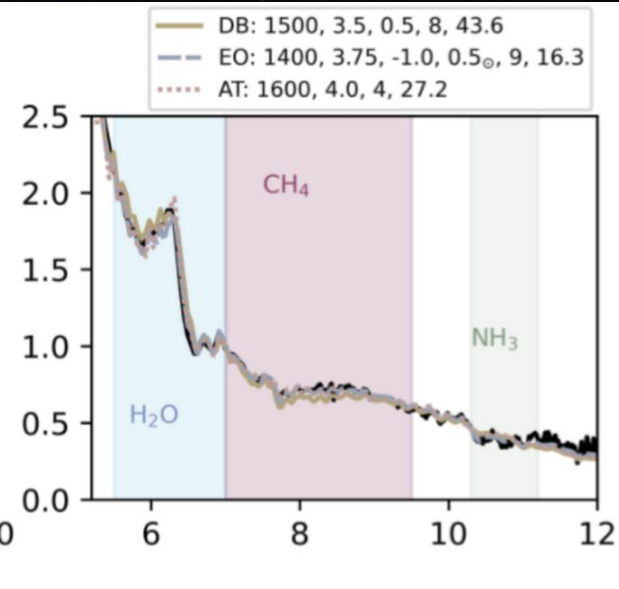
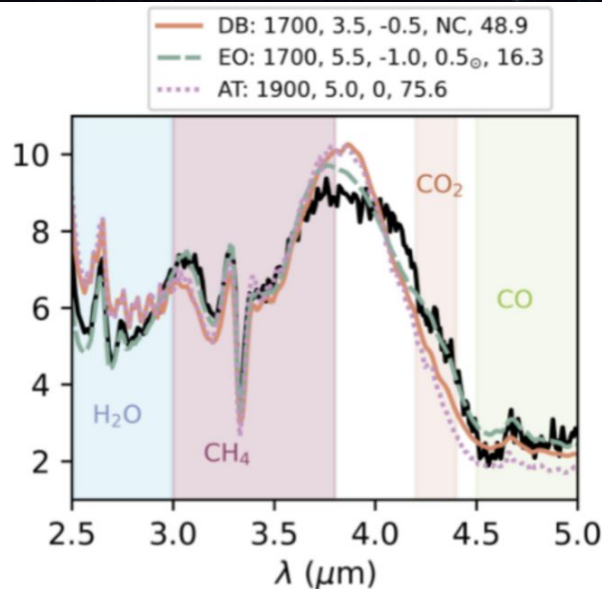
Rothermich et al. (2026, submitted)

Application 1: Modeling Multiple Wavelength Regions

Best fit: Cloudy models



Best fit: Cloud-free models



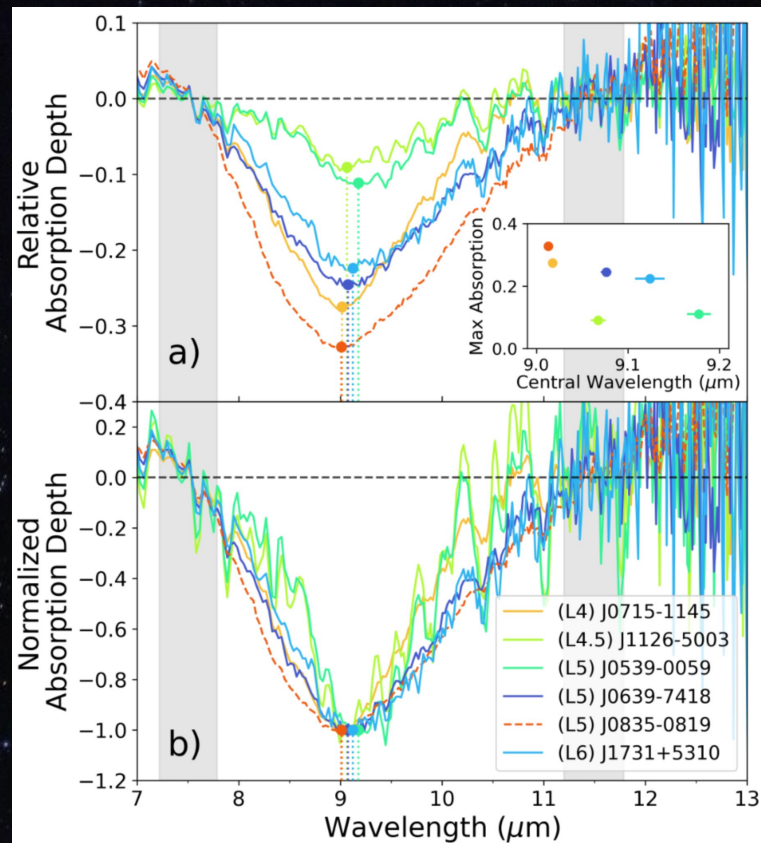
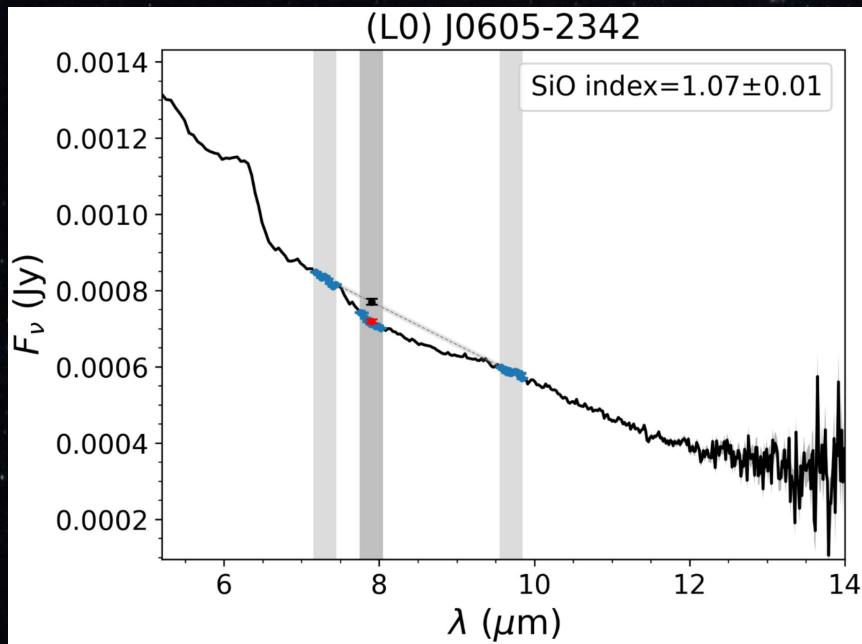
Rothermich et al. (2026, submitted)

Evidence of cloud sedimentation!

Application 2: Extracting Absorption Profiles

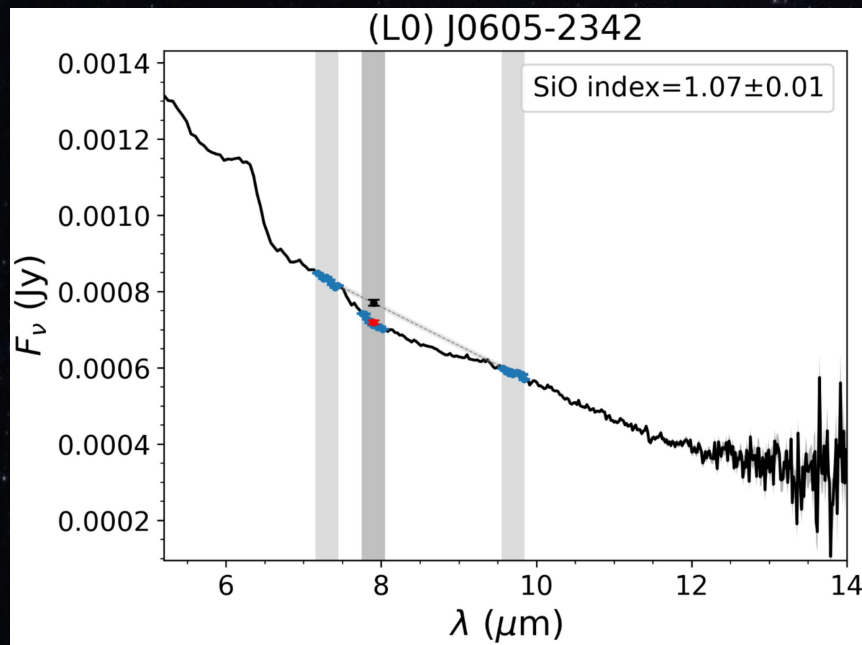


Application 2: Extracting Absorption Profiles

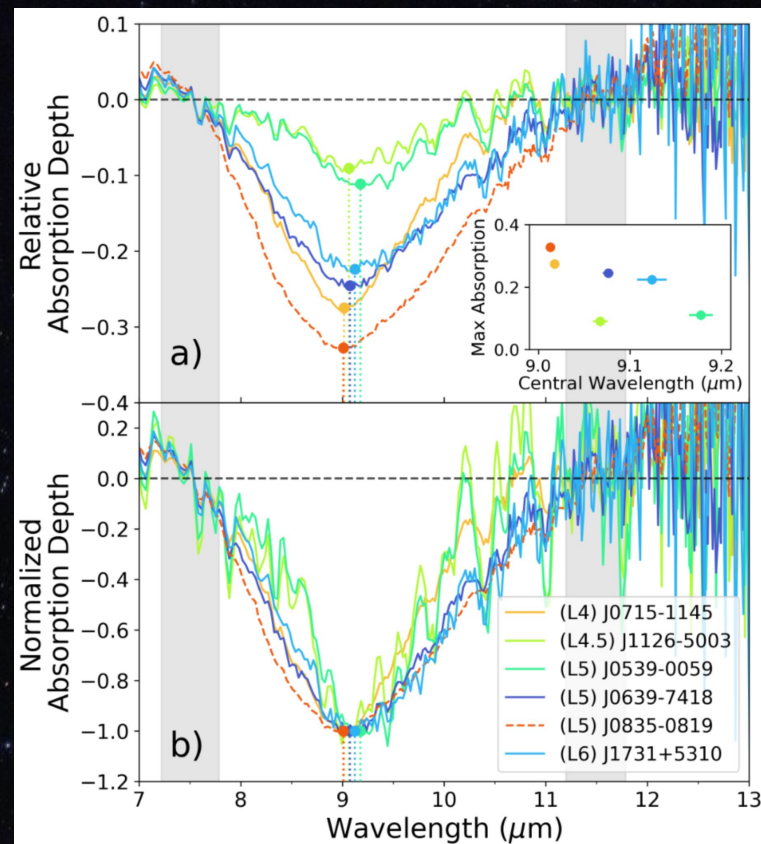


Barber, Metchev, Suárez et al. (2026, ~submitted)

Application 2: Extracting Absorption Profiles

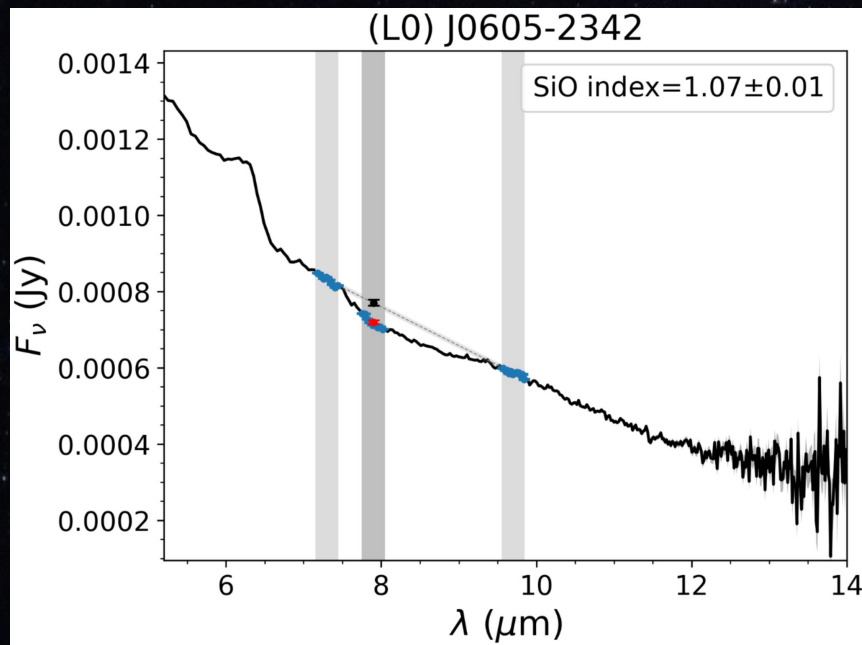


First detection of SiO gas in substellar atmospheres.



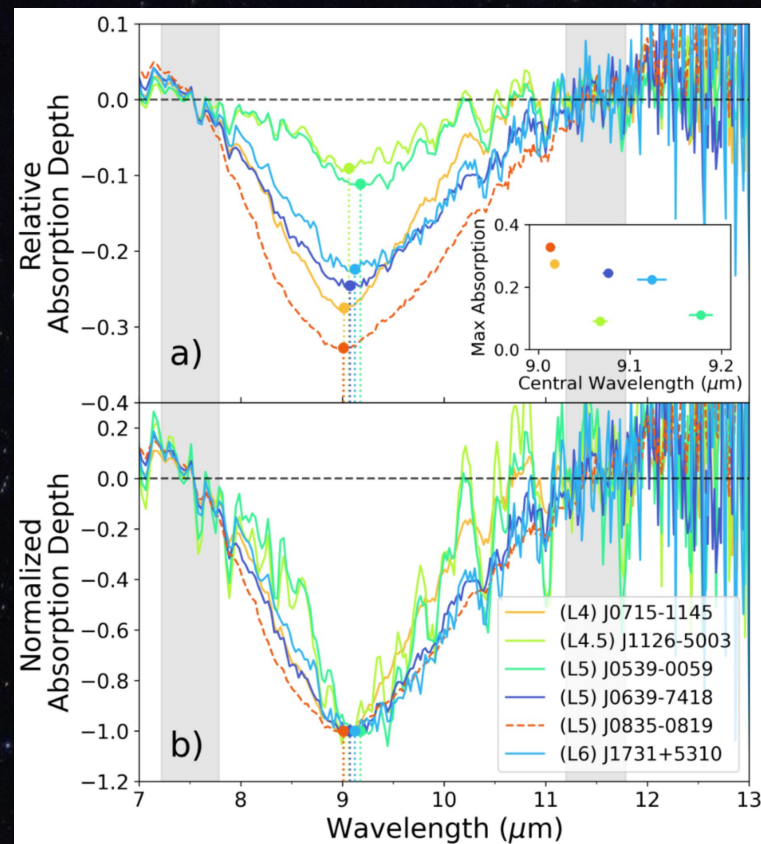
Barber, Metchev, Suárez et al. (2026, ~submitted)

Application 2: Extracting Absorption Profiles



First detection of SiO gas in substellar atmospheres.

Tentative correlation between silicate absorption depth and the wavelength of maximum depth.

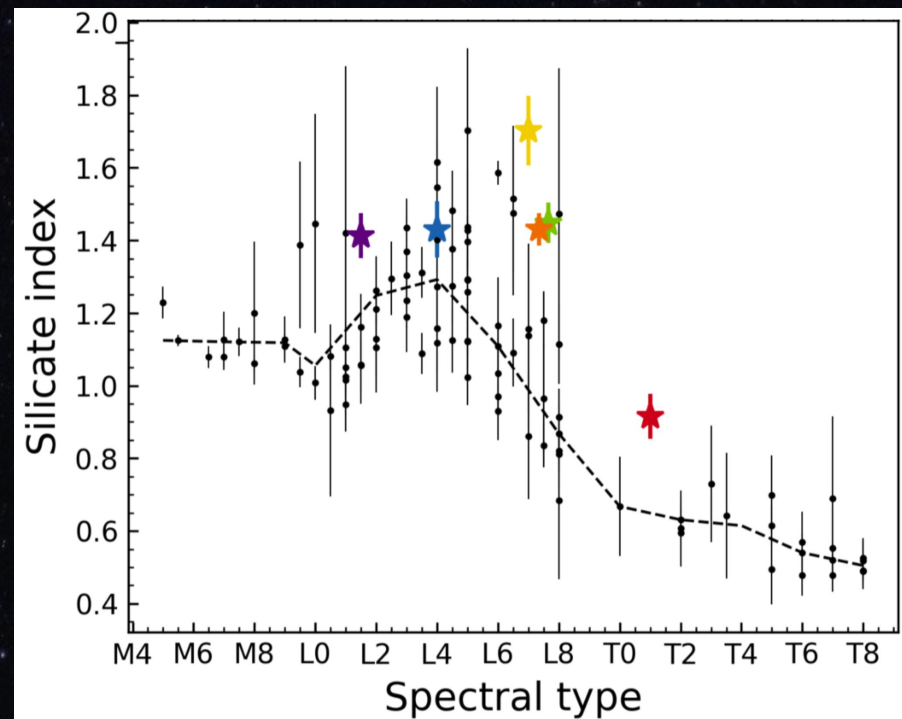
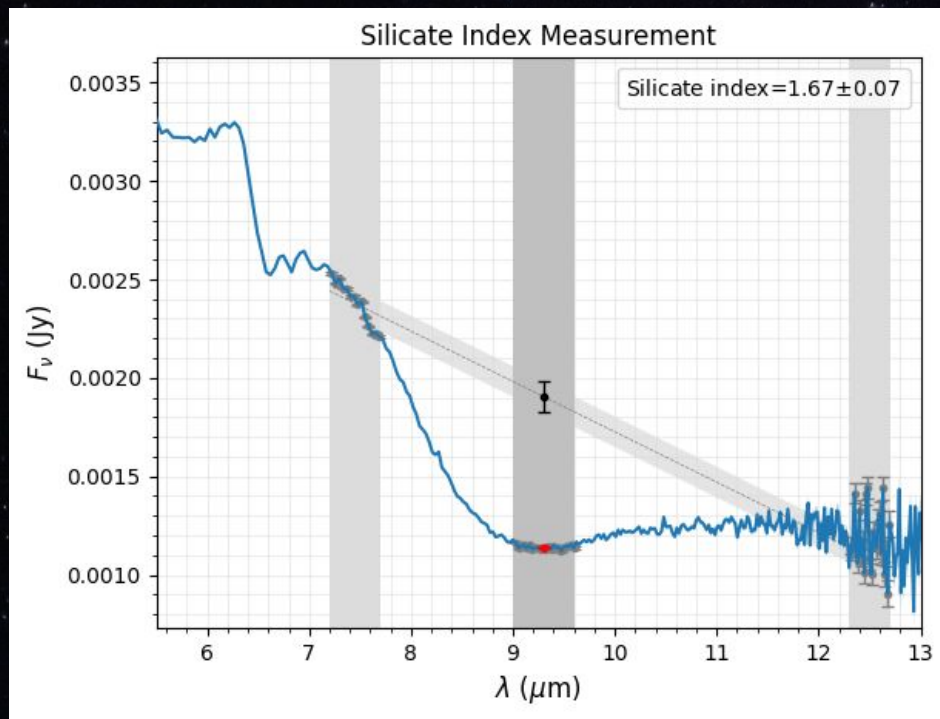


Barber, Metchev, Suárez et al. (2026, ~submitted)

Application 3: Silicate Depth Analysis Across Different Parameters

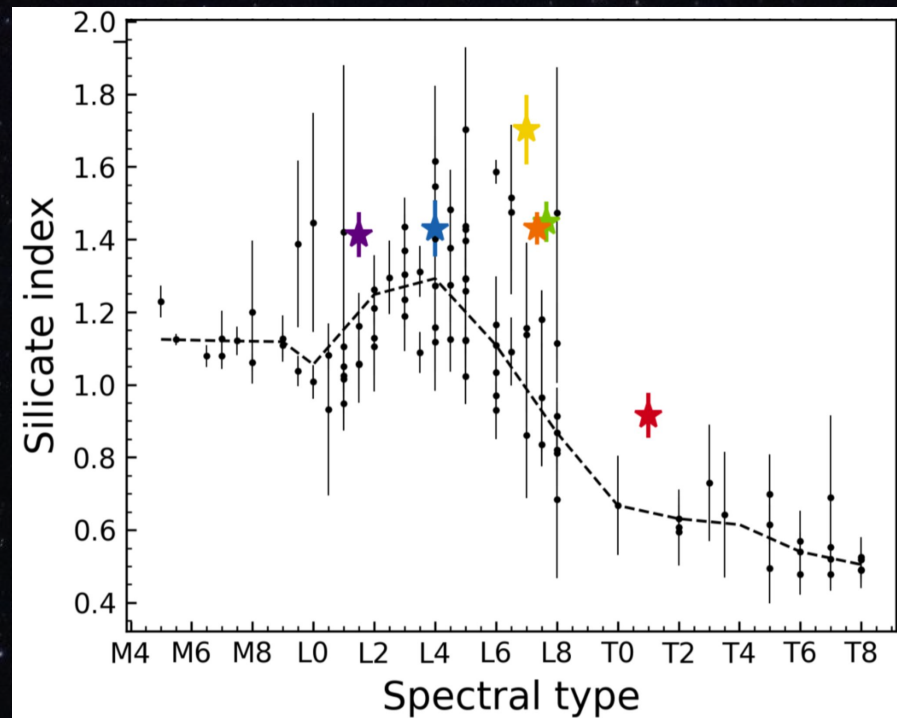
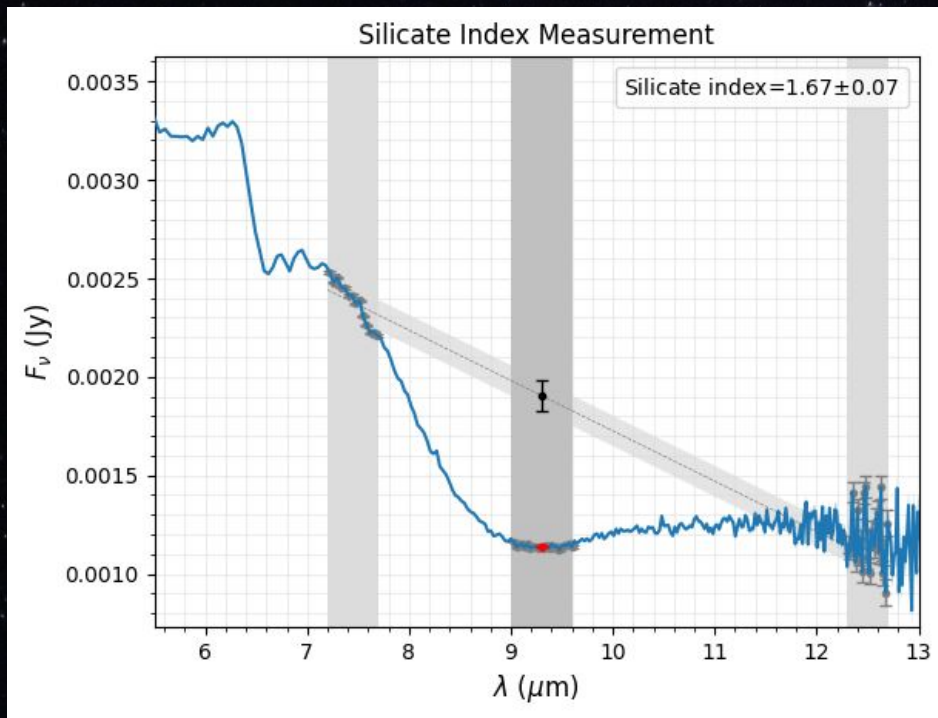


Application 3: Silicate Depth Analysis Across Different Parameters



Lam, Vos, Suárez et al. (2026)

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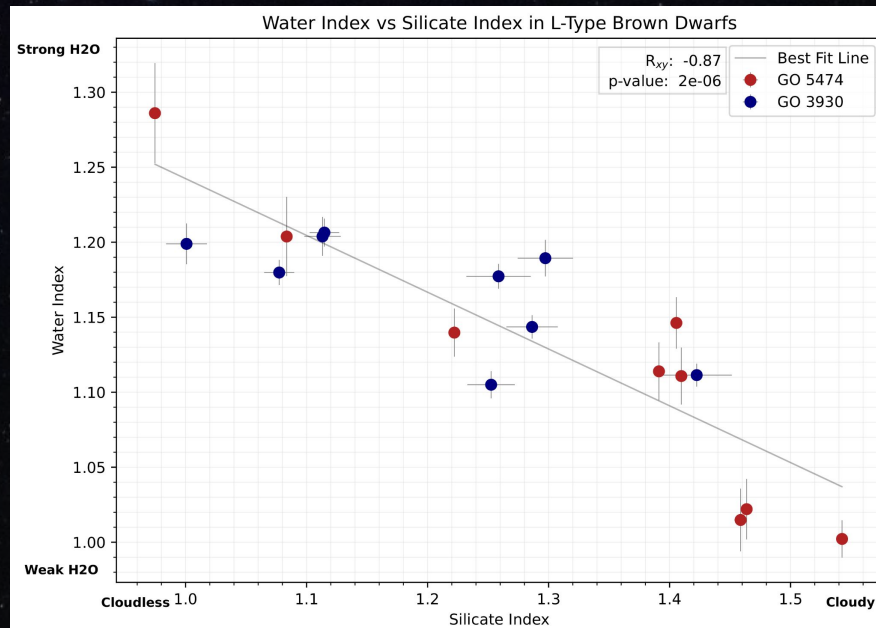
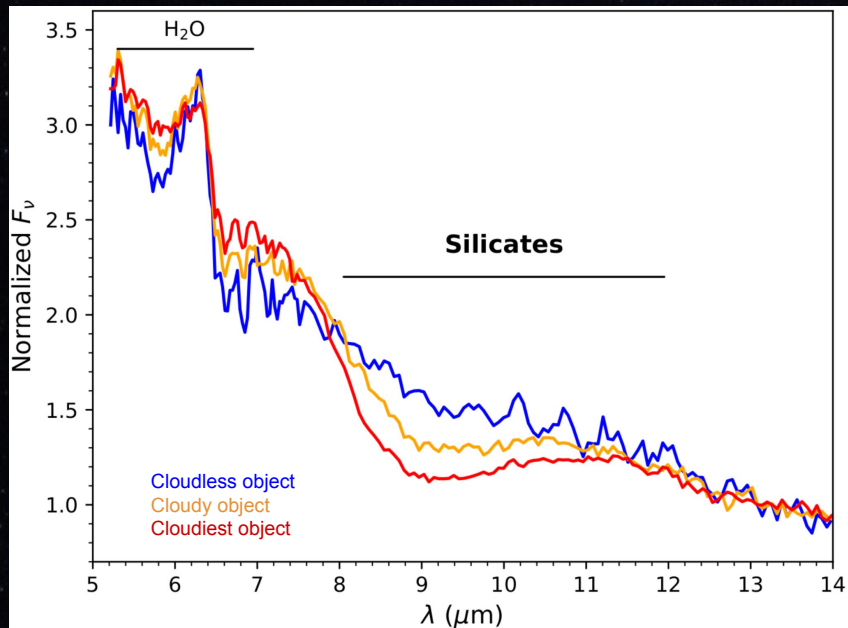
Lam, Vos, Suárez et al. (2026)

Discovery of the strongest silicate absorption ever observed.

Application 4: Impact of Silicate Clouds on Emergent Spectra

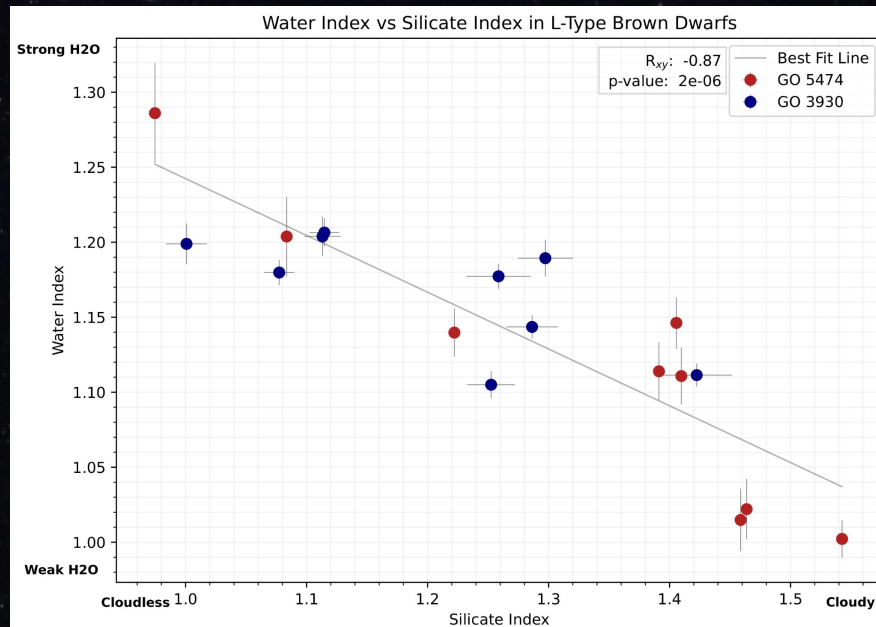
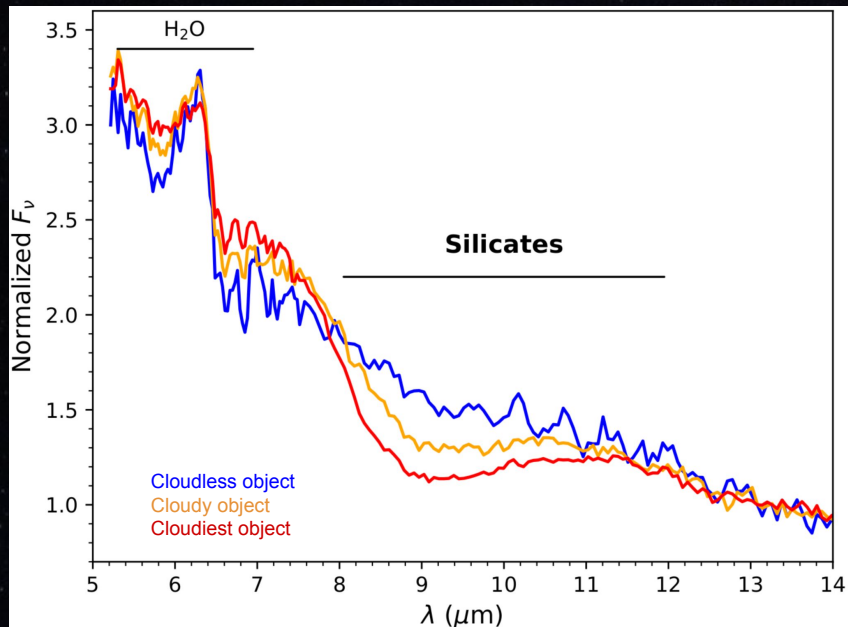


Application 4: Impact of Silicate Clouds on Emergent Spectra



Candero, Suárez et al. (2026, in prep.)

Application 4: Impact of Silicate Clouds on Emergent Spectra



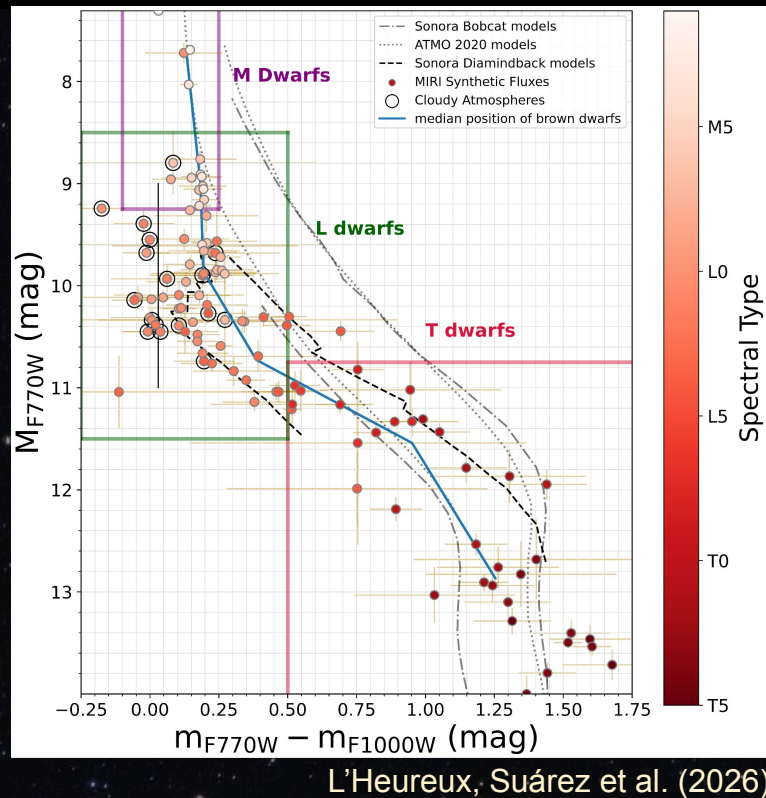
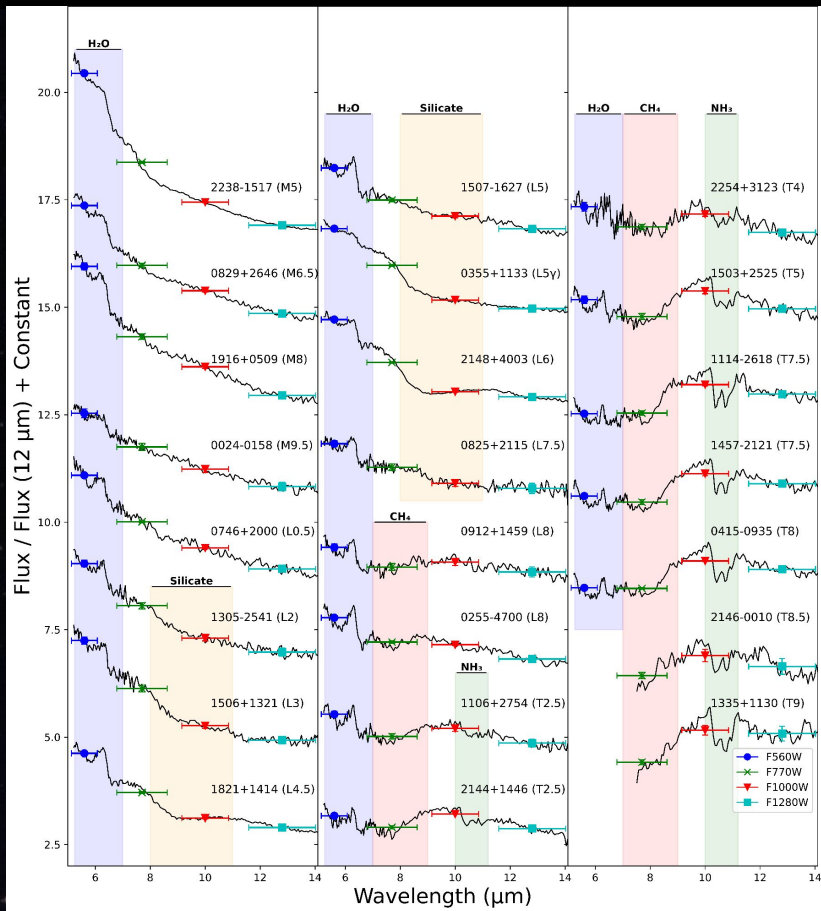
Candero, Suárez et al. (2026, in prep.)

Strong anticorrelation between silicate and water absorption depths!

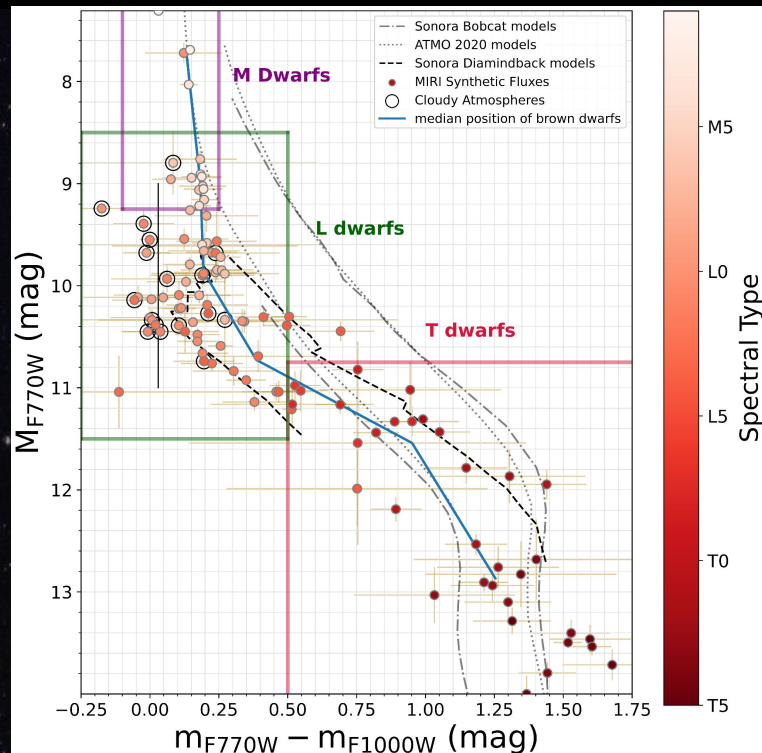
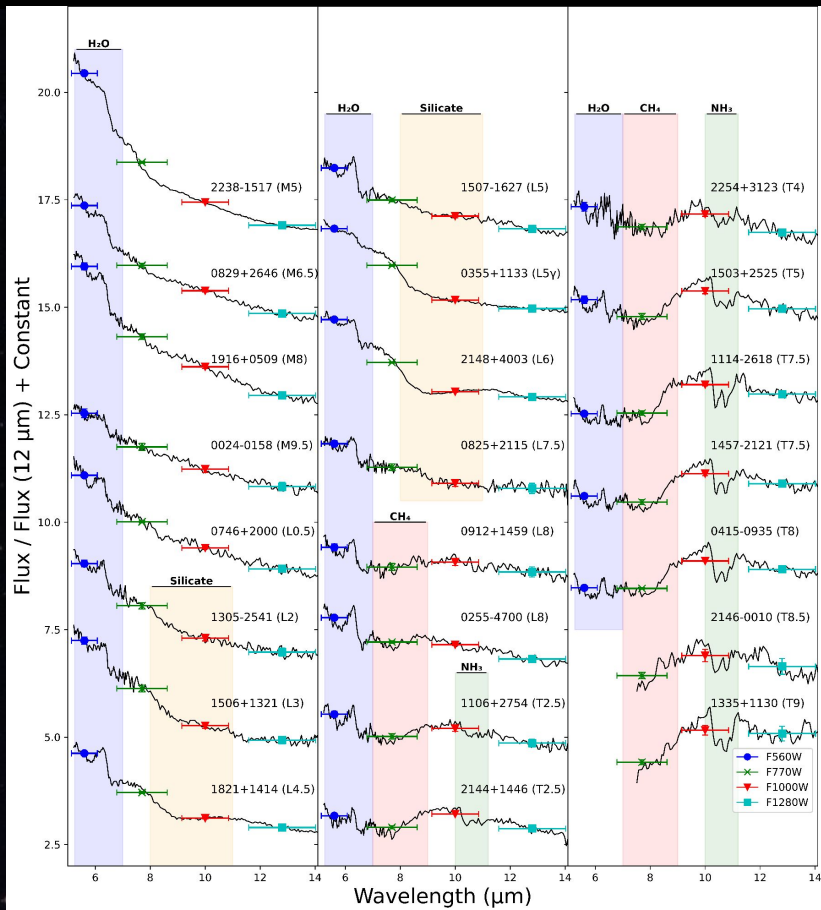
Application 5: Detecting Clouds with JWST/MIRI Imaging



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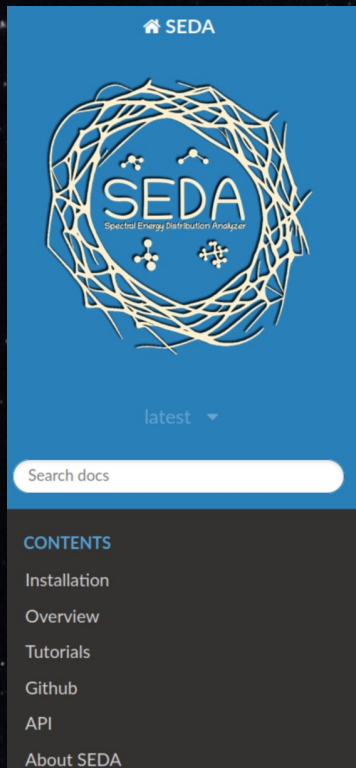


L'Heureux, Suárez et al. (2026)

Cloudy substellar atmospheres occupy distinct regions in MIRI color-magnitude diagrams!

Takeaway

Check out the SEDA python package to analyze SEDs of ultracool objects.



SEDA: Spectral Energy Distribution Analyzer

SEDA is an open-source Python package for forward modeling and empirical analysis of spectral energy distributions for ultracool objects, including brown dwarfs, directly imaged exoplanets, and low-mass stars. The code compares observed spectrophotometric data to atmospheric models by using a Bayesian framework to sample posteriors. Alternatively, the code minimizes the chi-square statistic to find the best model fits. SEDA also includes several tools useful for the analysis of spectral energy distributions and a variety of functions to visualize atmospheric models and results.

Check out the content below for further information on how to install and use SEDA.

Contents

- Installation
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 - Developer Installation
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 - Code Workflow of Forward Modeling
 - Useful Tools
 - Available Atmospheric Models

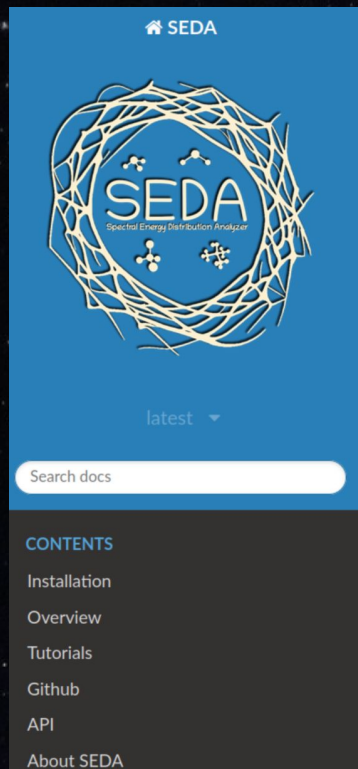


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 - Handling Atmospheric Model Grids
 - Chi-square minimization
 - Bayesian sampling
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 - Attribution
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Suárez et al. (2026, JOSS, submitted)

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Thanks for listening!

Suárez et al. (2026, JOSS, submitted)